



Society of Petroleum Engineers

SPE-229832-MS

Accelerating Field Development Planning with CF ++: A Petronas Digitalization Success Story

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229832-MS](https://doi.org/10.2118/229832-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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The oil and gas industry are undergoing significant transformation driven by digital technologies, sustainability imperatives, and the need for operational efficiency. In the upstream sector, Front-End project delivery has traditionally relied on manual workflows, fragmented data sources and workstreams, and siloed decision-making processes. These limitations often result in delays, cost overruns, and suboptimal investment decisions.

Recognizing these challenges, PETRONAS initiated a digitalization strategy aimed at enhancing its Front-End processes. A key outcome of this strategy is the development of Concept Factory++ (CF++), a digital application designed to streamline Screening and Feasibility studies. CF++ integrates technical and cost analytics, benchmarking, and data connectivity to enable early, accurate concept definition and cost estimation.

This paper explores the development, deployment, and impact of CF++ within PETRONAS's upstream operations. It provides a comprehensive overview of the tool's objectives, design methodology, implementation process, and performance outcomes, demonstrating how CF++ has transformed Front-End project delivery.

Introduction

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Recognizing these challenges, PETRONAS initiated a digitalization strategy aimed at enhancing its Front-End processes. A key outcome of this strategy is the development of Concept Factory++ (CF++), a digital application designed to streamline Screening and Feasibility studies. CF++ integrates technical and cost analytics, benchmarking, and data connectivity to enable early, accurate concept definition and cost estimation.

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implementation process, and performance outcomes, demonstrating how CF++ has transformed Front-End project delivery.

Objective and Scope

Concept Factory++ (CF++) is PETRONAS's flagship field development planning application, developed and deployed as part of its Front-End digitalization initiative since early 2021. The primary objective of CF++ is to enhance the quality, speed, and consistency of Front-End project delivery by replacing manual workflows with a fully digital, user-centric solution.

Built using Agile development methodology, CF++ was designed to be iterative, adaptable, and responsive to evolving business needs. Its scope encompasses Screening and Feasibility studies, with seamless integration into the broader Field Development Planning (FDP) workspace. This ensures alignment with PETRONAS's strategic goals of enabling faster, smarter, and more collaborative project execution.

Since its deployment, CF++ has been actively used across multiple upstream projects, delivering measurable improvements in technical definition, cost estimation accuracy, and decision-making confidence. Compared to conventional methods, CF++ has significantly elevated user experience, reduced cycle times, and improved the overall quality of early-stage project evaluations.

By embedding digital intelligence into the Front-End workflow, CF++ supports PETRONAS's ambition to drive systemic shifts in Facilities Engineering and project planning, in line with the strategic shifts outlined during Action Day and endorsed by leadership.

Methodology

Development Approach and Process

Concept Factory++ (CF++) was developed by PETRONAS using Agile methodology, emphasizing iterative development, continuous feedback, and active user involvement. This approach enabled rapid prototyping, testing, and refinement of the application based on real-world use cases and evolving business needs. The recent completion of Release 4 exemplifies this agility, delivering enhanced functionality and user experience improvements that support the full spectrum of offshore and onshore development studies.

The transformation to CF++ was driven by the need to overcome longstanding challenges in Front-End project planning—namely, manual inefficiencies, fragmented data silos, and inconsistent outputs that hindered speed, quality, and confidence in decision-making. CF++ addresses these challenges through a suite of integrated features that collectively redefine how early-stage development is executed:

Key features of CF++ include:

- **Auto-Generated Viable Concept Generation**

CF++ uses a robust logic and rules-based technical engine to automatically generate all viable facility development concepts based on planning input parameters. This ensures comprehensive screening and eliminates gaps that could lead to sub-optimal concept selection and value leakage.

- **Integrated Technical and Cost Analytics Engines**

These engines allow simultaneous evaluation of engineering feasibility and cost implications, ensuring alignment between technical and commercial considerations from the outset.

- **Embedded Benchmarking**

CF++ incorporates both industry and internal benchmarks to validate concept feasibility and cost estimates, enhancing credibility and reducing reliance on high contingencies.

- **Seamless Data Integration**

The tool connects with existing PETRONAS data sources and systems, eliminating silos and enabling real-time access to relevant information for faster, more informed decisions.

• User-Centric Design

With an intuitive interface and customizable workflows, CF++ supports diverse project requirements and enhances user experience across disciplines.

The development of Concept Factory++ (CF++) was anchored in Agile methodology, a strategic choice that allowed PETRONAS to respond dynamically to user needs, evolving business requirements, and technical challenges. This approach emphasized:

- **Iterative Development:** CF++ was built in progressive cycles, with each iteration delivering incremental improvements. This allowed the team to test features early, gather feedback, and refine functionality before full-scale deployment.
- **Continuous Feedback Loops:** Regular engagement with end-users—Facilities Engineers, planners, and project teams—ensured that CF++ remained aligned with real-world workflows. Feedback was actively incorporated into each sprint, driving enhancements in usability, accuracy, and relevance.
- **Cross-Functional Collaboration:** The development team included domain experts from engineering, cost estimation, data science, and software development. This multidisciplinary approach ensured that CF++ was technically robust, commercially sound, and operationally practical.
- **Rapid Prototyping and Testing:** Early versions of CF++ were deployed in controlled environments to validate core functionalities such as concept generation, cost analytics, and benchmarking. These prototypes helped identify gaps and opportunities for improvement before broader rollout.
- **Scalable Architecture:** CF++ was designed with scalability in mind, enabling seamless expansion from initial Screening studies to full Feasibility and Field Development Planning (FDP) integration. The architecture supports both offshore and onshore development scenarios.
- **Release Management:** Each release cycle was structured to deliver specific value enhancements. For example, Release 4 introduced:
 - Expanded functionality to cover a wider range of development types.
 - Enhanced user experience through interface redesign and workflow customization.
 - Improved cost granularity and benchmarking capabilities.
 - Greater integration with PETRONAS's digital ecosystem, enabling real-time data access and collaboration.

The current state of CF++ represents a significant leap in value creation, transitioning from manual workflows to a fully digital environment. This transformation has enabled seamless integration into the Field Development Planning (FDP) workspace, fostering more collaborative, data-driven decision-making. With enhanced cost granularity and improved accuracy in technical definition and estimation, CF++ empowers project teams to make faster, smarter choices with greater confidence, directly supporting PETRONAS's strategic goals for upstream excellence.

A summary flowchart depicting CF++ architecture is shown in [Figure 1](#). The architecture of Concept Factory++ (CF++) is designed to support end-to-end digitalization of Front-End (FE) project planning, integrating data, automation, and analytics to enable faster, smarter decision-making. It consists of interconnected modules that work together to streamline concept generation, cost estimation, and project evaluation. This modular and scalable architecture enables CF++ to support the full lifecycle of Front-End planning, from initial screening to feasibility studies, while ensuring consistency, transparency, and strategic alignment across PETRONAS's upstream portfolio.

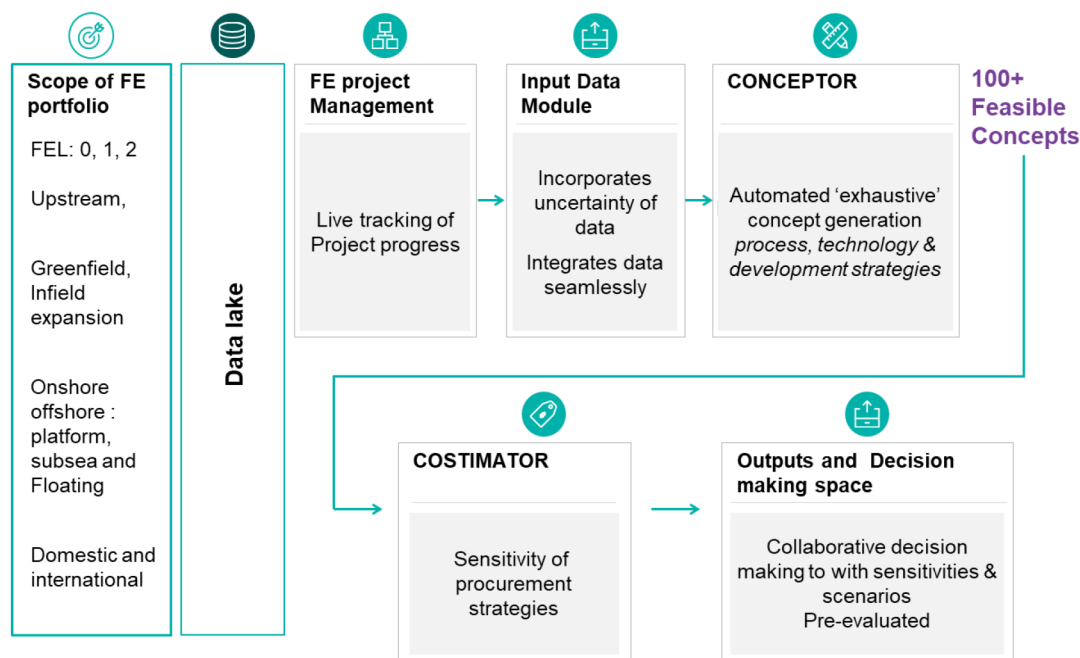


Figure 1—CF++ Architecture

Implementation Process

The implementation of Concept Factory++ (CF++) was a structured and collaborative effort, designed to ensure the solution delivered tangible improvements in Front-End planning while aligning with PETRONAS's broader digital transformation goals. The process involved several key phases:

- i. **Needs Assessment:** PETRONAS began by conducting a comprehensive review of existing Front-End workflows, identifying critical pain points such as manual inefficiencies, data fragmentation, and inconsistent outputs. This diagnostic phase helped define the scope of transformation and prioritize areas where digital intervention could deliver the greatest value.
- ii. **Stakeholder Engagement:** A cross-functional engagement strategy was deployed to ensure alignment across technical, commercial, and operational domains. Stakeholders—including Facilities Engineers, planners, cost estimators, and IT specialists—were involved in:
 - Defining functional and technical requirements
 - Validating early prototypes
 - Providing feedback to shape the user experience and ensure business relevance

This inclusive approach fostered ownership and accelerated adoption.

- iii. **System Integration:** CF++ was integrated with PETRONAS's existing digital platforms and data ecosystems, including project databases, cost repositories, and benchmarking tools. This enabled:
 - Seamless data flow across modules
 - Real-time collaboration between teams
 - Elimination of silos and duplication

The integration ensured CF++ could operate as a central hub within the broader Field Development Planning (FDP) workspace.

- iv. **Training and Change Management:** Recognizing the importance of user readiness, PETRONAS launched a comprehensive training and change management program. This included:
 - o Hands-on training sessions tailored to different user groups
 - o Digital learning resources and user guides
 - o Change champions embedded within teams to support onboarding and troubleshooting

These initiatives helped build confidence and competence in using CF++, accelerating its adoption across the organization.

- v. **Continuous Improvement:** In line with Agile development principles, PETRONAS established feedback loops to capture user insights and drive ongoing enhancements. Functionality was scaled progressively based on:
 - o Portfolio needs
 - o User feedback
 - o Value delivered within the Front-End work process

This iterative approach ensured CF++ remained responsive to evolving business requirements and continued to deliver measurable improvements in speed, quality, and decision-making confidence.

Results and Observations

Performance Gains

The deployment of Concept Factory++ (CF++) has delivered substantial performance gains across PETRONAS's Front-End planning processes. A comparative analysis against traditional workflows revealed improvements in speed, quality, collaboration, and efficiency, validating the strategic value of digital transformation.

- **Speed:** CF++ significantly accelerated project delivery timelines by automating concept generation and cost estimation, reducing manual effort, and streamlining decision workflows. Early-stage evaluations that previously took weeks can now be completed in days.
- **Quality:** The integration of technical and cost analytics engines, coupled with embedded benchmarking, resulted in higher fidelity outputs. Concepts are now defined with greater precision, and cost estimates are more robust and defensible, reducing reliance on high contingencies.
- **Collaboration:** Cross-functional teams reported improved alignment and transparency. The centralized platform and shared data environment enabled more effective collaboration between engineering, planning, and commercial teams, fostering consensus and reducing iteration cycles.
- **Efficiency:** CF++ reduced rework by providing clearer decision gates and more accurate early-stage data. This allowed smoother progression through the project funnel, minimizing delays and resource wastage.

Validation and Verification

To ensure the reliability and robustness of CF++, PETRONAS conducted a series of rigorous validation and verification exercises using real project data across various upstream scenarios. These exercises were designed to test the tool's performance against actual outcomes and industry standards, confirming its consistency, accuracy, and alignment with techno-commercial benchmarks.

One of the key findings was the accuracy of CF++ in generating cost estimates. The tool consistently produced outputs within acceptable variance ranges when compared to actual project results, demonstrating its capability to support early-stage decision-making with credible data. Additionally, CF++ offered strong traceability, with clear audit trails and documentation embedded throughout the workflow. This enhanced governance and accountability, allowing users to track assumptions, inputs, and changes across the planning lifecycle.

User satisfaction was another critical indicator of success. Feedback gathered from project teams highlighted high levels of confidence in the tool's usability, functionality, and impact. Users appreciated the intuitive interface, the transparency of the analytics, and the ability to collaborate more effectively across disciplines. These validation outcomes reinforced CF++ as a trusted and scalable solution for Front-End planning, capable of delivering consistent value across PETRONAS's upstream portfolio.

Discussion

Addressing Legacy Inefficiencies

Traditional Front-End (FE) workflows in the upstream oil and gas sector have long been hindered by manual data handling, fragmented systems, and inconsistent outputs. These legacy inefficiencies often result in delayed project timelines, increased costs, and suboptimal decision-making due to limited visibility and disconnected processes.

Concept Factory++ (CF++) was purposefully designed to address these systemic challenges by introducing a fully digital, integrated, and intelligent planning environment. By automating technical and cost evaluations, CF++ eliminates the need for repetitive manual calculations and reduces human error. Its ability to provide real-time access to integrated data ensures that all stakeholders are working from a single source of truth, improving transparency and reducing misalignment.

Furthermore, CF++ embeds benchmarking capabilities that allow users to validate outputs against internal and industry standards, enhancing the credibility of early-stage estimates and reducing reliance on high contingencies. The platform also fosters cross-functional collaboration through shared digital workspaces, enabling engineering, planning, and commercial teams to co-develop and evaluate concepts in a unified environment.

Collectively, these capabilities represent a significant departure from traditional practices, enabling PETRONAS to modernize its Front-End planning approach and unlock greater speed, accuracy, and value in upstream project development.

Strategic and Sustainability Impact

Beyond operational improvements, Concept Factory++ (CF++) plays a strategic role in advancing PETRONAS's broader goals, particularly in the areas of resource optimization, sustainability, and investment performance. By enabling faster, more accurate, and data-driven project evaluations, CF++ supports better capital allocation and portfolio prioritization, ensuring that resources are directed toward the most viable and value-accretive opportunities.

From a sustainability perspective, CF++ contributes to reducing environmental impact by facilitating early identification of lower-emission development options. The tool's evolving capabilities now include the early determination of Carbon Footprint and Carbon Intensity as part of comparative concept assessments, allowing environmental considerations to be embedded into decision-making from the outset.

Moreover, the platform is continuously enhanced to remain aligned with evolving business priorities, regulatory expectations, and industry standards. This ensures that CF++ not only addresses current operational needs but also positions PETRONAS to lead in responsible and future-ready upstream development.

Benchmarking Against Industry Standards

CF++ stands out as one of the few fully integrated digital solutions in the upstream sector, setting a new benchmark for digitalization in Front-End project planning. Its comprehensive capabilities—spanning automated concept generation, cost estimation, benchmarking, and collaborative decision-making—position PETRONAS as a leader in digital project delivery. Comparative benchmarking exercises have demonstrated that CF++ consistently outperforms traditional tools in key performance areas, including speed, accuracy, and user experience. By unifying technical and commercial workflows within a single platform, CF++ not only enhances operational efficiency but also reinforces PETRONAS's commitment to innovation and excellence in upstream development.

Project A: Offshore Development Screening Case Study

The project is a gas development initiative evaluated under a structured field development framework. Initially, the framework generated multiple conceptual options for the field, and through expert judgment, one concept was selected for further development. This selected concept is designed with a total project duration of over five years.

The capital expenditure (CAPEX) estimate is based on a preliminary Class 5 estimate (AACE), incorporating a wide accuracy range and a significant cost contingency to account for uncertainties typical at this stage.

The field development is straightforward and lean, tailored for gas production with minimal infrastructure complexity. Based on engineering input, the framework has defined the following parameters for the selected concept:

- Well Architecture: Two producer wells, no injector wells, and one drill centre.
- Processing Philosophy: No processing at the wellhead. The system includes essential components such as the wellhead, control modules, production header, and tie-in point.
- Conceptual Layout: A subsea wellhead platform (WHP) is connected to an existing WHP via a flowline and pipeline, forming the backbone of the production route.

The cost structure is competitive, with CAPEX and operational expenditure (OPEX) benchmarked against industry standards. The CAPEX falls within a reasonable range of the industrial average, indicating strong cost competitiveness.

Regarding injection and disposal strategy, the project does not utilize water disposal or non-associated gas (NAG) reservoirs. There is no nearby gas field available for injection, and the available gas supply is minimal with high operating pressure, making reinjection infeasible.

The evacuation scheme is designed as a tie-back to existing facilities, using a carbon steel pipeline spanning several kilometers from the drill centre to the evacuation point. The pipeline is appropriately sized and designated for service.

Overall, the selected concept demonstrates a lean, cost-effective, and operationally sound development model. It is particularly suited for gas fields with limited processing and disposal options. By leveraging existing infrastructure and a simplified evacuation strategy, the project minimizes capital investment while maintaining technical and operational integrity. The sample of CF++ User Interfaces are shown in [Figure 2](#), [3](#), [4](#), [5](#) and [6](#).

DATA INPUT CONCEPT DETAILS SCHEDULE CFA CARBON ABATEMENT

Input Summary

FIELD DATA	
Hydrocarbon Type	Gas
Production Year	2027
Production Life	10 years
Abandonment Pressure	500 PSIA
Total Recoverable Reserve (PSO)	58.96
Availability of Water Disposal Reservoir	No
Water Disposal Location	0
Availability of NAG Reservoir For Gas Injection/Lift	No
Availability of Nearby Gas Field For Gas Injection/Lift	No
Available Amount of Gas To Be Supplied	0 MMscfd
Operating Pressure	30 bar
Gas Disposal by Reinjection	No

ENVIRONMENTAL DATA	
AMBIENT TEMPERATURE (MIN)	25.82 °C
AMBIENT TEMPERATURE (MAX)	27.35 °C
SEABED TEMPERATURE (MIN)	15.6 °C
SEABED TEMPERATURE (MAX)	23.4 °C

Figure 2—CF++ Input Data

CONCEPT SELECTION 11 Total Concepts 1 BOOKMARKED CONCEPTS SAVE INPUTS

☒ Show all generated concepts
☒ Show Base concepts

PRODUCTION PROFILE: P50 P50 P50

HOST BY: CAPEX OPER UTE

SCHEDULE

Facility Design Choices

- PROCESSING SCHEME
- EVACUATION SCHEME
- TREE TYPE
- WELL TEST
- ACCOMMODATION
- DW STRATEGY
- CONDENSATE PROCESSING
- HOST TYPE

Concept 5 (Base Concept)

Overall: 45 months
Execution: 26 months

CAPEX: \$1.2M OPEX: \$0.1M

0.000 / 0.000 MMscfd 0.0 MM

Oil Prod: 0 Gas Prod: 0 Water Inj: 0

No Processing N/A

The back to Existing Facilities 0 suggested

Concept 4

Overall: 64 months
Execution: 26 months

CAPEX: \$1.2M OPEX: \$0.1M

0.000 / 0.000 MMscfd 0.0 MM

Oil Prod: 0 Gas Prod: 0 Water Inj: 0

No Processing N/A

The back to Existing Facilities 0 suggested

Concept 3

Overall: 80 months
Execution: 40 months

CAPEX: \$1.2M OPEX: \$0.1M

0.000 / 0.000 MMscfd 0.0 MM

Oil Prod: 0 Gas Prod: 0 Water Inj: 0

Full Processing N/A

The back to Existing Facilities 0 suggested

Concept 11

Overall: 40 months
Execution: 40 months

CAPEX: \$1.2M OPEX: \$0.1M

0.000 / 0.000 MMscfd 0.0 MM

Oil Prod: 0 Gas Prod: 0 Water Inj: 0

Full Processing N/A

The back to Existing Facilities 0 suggested

Concept 2

Overall: 80 months
Execution: 40 months

CAPEX: \$1.2M OPEX: \$0.1M

0.000 / 0.000 MMscfd 0.0 MM

Oil Prod: 0 Gas Prod: 0 Water Inj: 0

Full Processing N/A

The back to Existing Facilities 0 suggested

Concept 6

Overall: 80 months
Execution: 40 months

CAPEX: \$1.2M OPEX: \$0.1M

0.000 / 0.000 MMscfd 0.0 MM

Oil Prod: 0 Gas Prod: 0 Water Inj: 0

Full Processing N/A

The back to Existing Facilities 0 suggested

Figure 3—CF++ Sample Generated Concept Cards

← Back to Project Overview **Concept 5** Generated 8 August, 2025

Concept Summary

PROJECT OVERVIEW

DATE STARTED: 8 August, 2025

Case profile: P50

LOCATION

COST INPUTS

CAPEX (MM USD) OPEX (MM USD)

Schedule Risk Score

WELLS

No. of Producer: 2
No. of Injector: 0
No. of Drill Centres: 1

CONCEPT DIAGRAM

PROCESS SCHEMATICS

Processing Philosophy: No Processing

Figure 4—CF++ Sample Concept Details

WEIGHT AND POWER GENERATION ESTIMATE			
TOPSIDE			
	DRY WEIGHT ESTIMATE (te)	OPERATING WEIGHT ESTIMATE (te)	POWER LOAD (MW)
Manifold			
Production	49.12	117.90	0.00
Oil Processing			
Wet Gas Meter	0.75	1.80	0.00
Control and Communications			
WHP	6.25	6.25	0.00
Utilities			
Firefighting	1.56	2.65	0.01
Mechanical Handling	24.59	26.06	0.00
Chemical Injection & Storage	5.38	9.14	0.04
Other Utilities	5.75	9.78	0.04
Flare			
Flare System	2.25	2.25	0.02
Power			
Power Generation (Diesel / Gas Engine Generator)	1.63	1.71	0.00
Power Distribution	0.52	0.52	0.00
Emergency Power	2.35	2.47	0.00
Bulks			
Fabrication (Equipment)	1,162.65	1,162.65	0.00
Fabrication (Structural)	1,392.55	1,392.55	0.00
Fabrication (Piping)	54.75	54.75	0.00
Fabrication (Electrical)	18.13	18.13	0.00
Fabrication (Instruments)	20.99	20.99	0.00
Fabrication (Others)	9.43	9.43	0.00
Grand Total	2,758.65	2,839.03	0.11

Figure 5—CF++ Sample of Work Breakdown Structure

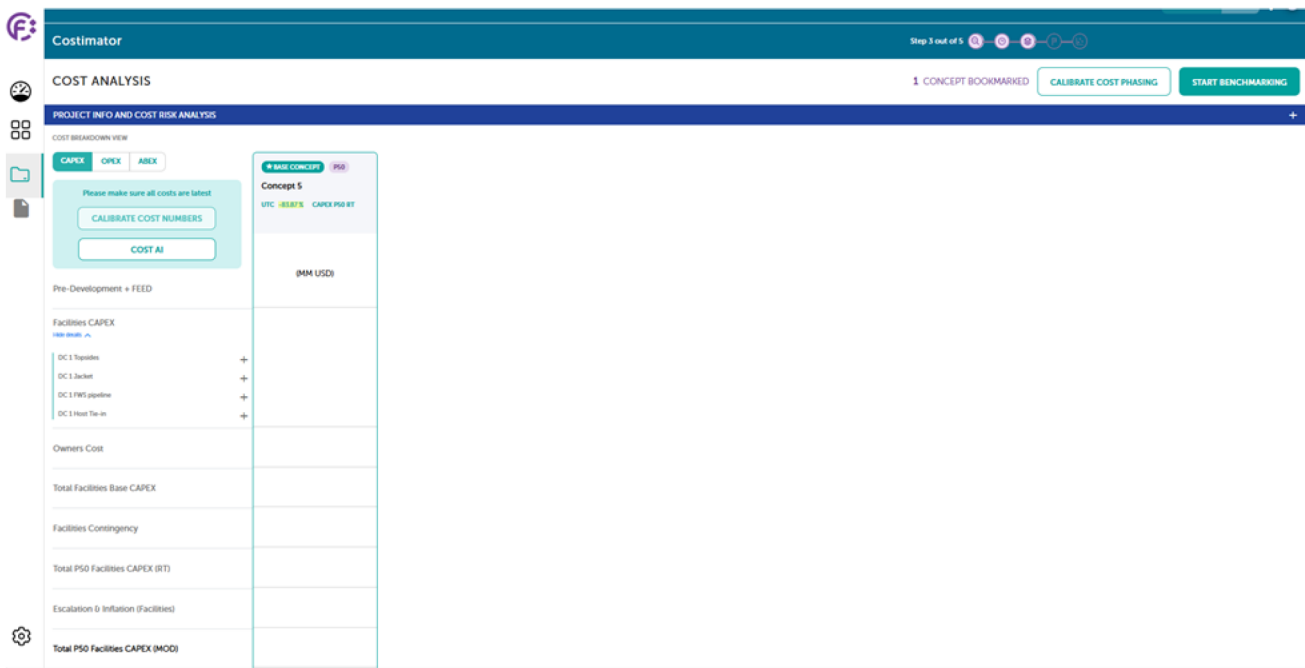


Figure 6—CF++ Sample of Cost Analysis Page

Driving Forward with CF++: Onshore Expansion, Offshore Enhancements and AI Integration

The Concept Factory++ (CF++) continues to evolve as a cornerstone digital platform for upstream facilities engineering, with the next release marking a pivotal phase in its transformation. This release targets to introduce a robust onshore module, designed to streamline project definition and accelerate front-end loading (FEL) deliverables, complimenting the existing offshore module.

On the offshore front, CF++ continues to build on its foundational modules with targeted enhancements. The Well Cost Phasing logic has been upgraded to support more granular cost breakdowns, while the Subsurface Input 2.0 improvements extend to offshore datasets. Newly reconfigured Conceptor engine that supports base concept generation is also part of the enhancement, together with the upgraded function

of schedule automation that allows for calibration flexibility in meeting specific projects' targets. These updates are critical for stabilising the Conceptor and Costimator modules, especially as CF++ scales to accommodate more complex offshore scenarios. The offshore module enhancements also support collaborative decision-making by enhancing the integration of subsurface, wells, and facilities data into a unified evaluation framework.

To elevate CF++ beyond its current rule-based capabilities, the platform is embarking on a strategic transformation into an AI-powered engine. This journey begins with a targeted Proof of Concept (PoC), designed to validate AI's potential in a controlled environment before scaling to full implementation. The PoC will focus on intelligent concept ranking and selection, leveraging multi-dimensional trade-off analysis across cost, schedule, emissions, and economics. It will also introduce automated database integration, enabling seamless access to historical benchmarks and real-time market data. This AI module is envisioned to be a game-changer by enhancing integration, boosting efficiency, and sharpening accuracy in FEL execution. It marks a decisive step towards smarter, faster, and more data-driven decision-making in upstream development.

Conclusion

Concept Factory++ (CF++) represents a transformative leap in PETRONAS's approach to Front-End project delivery, moving beyond incremental improvements to enable systemic change across the upstream planning landscape. By replacing manual, siloed workflows with a fully digital, integrated solution, CF++ enhances speed, quality, and collaboration throughout Screening and Feasibility studies. Its embedded analytics, benchmarking capabilities, and seamless data connectivity empower teams to define concepts and estimate costs with greater accuracy and confidence—reducing contingencies and enabling more informed, timely decisions.

The tool has delivered measurable value across multiple projects, validated through comparative performance analysis and strong user feedback. More importantly, CF++ serves as a key enabler of the systemic shifts outlined in PETRONAS's strategic agenda.

As PETRONAS continues to enhance CF++ through iterative development, the tool is poised to play an increasingly critical role in enabling faster, smarter, and more collaborative project delivery. CF++ sets a new benchmark for digitalization in Front-End workflows, offering a scalable, adaptable, and future-ready solution to meet the complex challenges of modern energy development.